

Hausdorff Dimensions of Quasi-Fuchsian Limit Sets Constructed via Schottky Groups

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Möbius map normalisation

- $M(z) = \frac{m_1 z + m_2}{m_3 z + m_4}, \quad M = \begin{pmatrix} m_1 & m_2 \\ m_3 & m_4 \end{pmatrix}$
- Homeomorphism $\phi : PSL(2, \mathbb{C}) \rightarrow \mathcal{M}$ s.t. $\phi \begin{pmatrix} m_1 & m_2 \\ m_3 & m_4 \end{pmatrix} = M(z)$
- Normalised eigenvector matrix $\phi(\hat{P}) = \phi \begin{pmatrix} \hat{v}_1 & \hat{v}_2 \end{pmatrix} \in \mathcal{M}$
- $\lambda_1 \neq \lambda_2 \implies \hat{P}^{-1} M \hat{P} = \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix}$
 - ▶ $\phi \begin{pmatrix} \lambda_1 & 0 \\ 0 & \lambda_2 \end{pmatrix} = \frac{\lambda_1}{\lambda_2} z$
- $\lambda_1 = \lambda_2 \implies \hat{P}^{-1} M \hat{P} = \begin{pmatrix} \lambda_1 & 1 \\ 0 & \lambda_1 \end{pmatrix} = \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix}$
 - ▶ $\phi \begin{pmatrix} 1 & 1 \\ 0 & 1 \end{pmatrix} = z + 1$

Classification of Möbius maps

- $k = \frac{\lambda_1}{\lambda_2}$
- $k + k^{-1} = \text{Tr}(M)^2 - 2$

Map Type	Multiplier	Trace	General Map Dynamics
Parabolic	$k = 1$	$\text{Tr}(M) = 2$	$S(z) = z + \alpha$
Elliptic	$k = e^{i\theta} \neq 1$	$ \text{Tr}(M) < 2$	$S(z) = e^{i\theta} z$
Hyperbolic	$1 \neq k > 0$	$\text{Tr}(M) > 2$	$S(z) = kz$
Loxodromic	$1 \neq k \notin \mathbb{R}$	$ \text{Tr}(M) > 2$	$S(z) = re^{i\theta} z$

Table: Map classifications

Isometric circles

- $M'(z) = \frac{1}{(m_3z + m_4)^2}$
- Derivatives describe scaling in local area
- $|M'(z)| = 1$ if $z \in \left\{ -\frac{m_4}{m_3} + \frac{e^{i\theta}}{|m_3|} \right\}$
 - ▶ Isometric circle of center $-\frac{m_4}{m_3}$ and radius $\frac{1}{|m_3|}$
- Inverse map $M^{-1}(z)$ has isometric circle $\left\{ -\frac{m_1}{m_3} + \frac{e^{i\theta}}{|m_3|} \right\}$
- $M(z)$ maps outside of its isometric circle inside the circle of $M^{-1}(z)$

Schottky groups

- Choose closed disks D_1, D_2, D'_1, D'_2
- Choose non-elliptic maps M_1, M_2
- Isometric circle of $M_i(z)$ bounds D'_i
- Isometric circle of $M_i^{-1}(z)$ bounds D_i
- A Schottky group is generated by maps M_1, M_2 with their associated Schottky disks
- $M_i(\hat{\mathbb{C}} \setminus D'_i) \subset D_i$

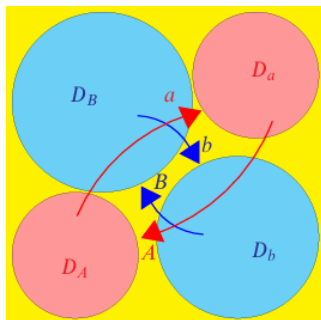


Image source: Indra's Pearls - David Mumford et al. (p.98)

Constructing limit sets

- Schottky group elements described by words of the generators
- Limit points are associated with words of infinite length
- The collection of all limit points is a limit set
- To construct images via code:
 - ▶ Use Depth First Search to collect words
 - ▶ Apply the word to a fixed point or Schottky disk
 - ▶ Plot the point or disk using Pyplot
- If limit set is connected, draw lines between nearby points
- If limit set is disconnected, set a limit to word length

Hausdorff measure & dimension

- $\mathcal{H}^d(X) = \lim_{\delta \rightarrow 0} \inf \left\{ \sum \text{diam}(B_i)^d : \text{diam}(B_i) < \delta, X \subseteq \bigcup B_i \right\}$
- All Borel sets are $\mathcal{H}^d$ -measurable
- $t > d \implies \mathcal{H}^t(X) \leq \lim_{\delta \rightarrow 0} \delta^{t-d} \mathcal{H}^d(X)$
- $\dim_H(X) = \inf \{d : \mathcal{H}^d(X) = 0\} = \sup \{d : \mathcal{H}^d(X) = \infty\}$

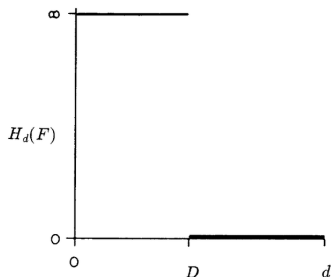


Figure: Graph showing Hausdorff measure of variable dimension. $D = \dim_H(X)$

Classical Schottky dust

- Choose generating Möbius maps to be hyperbolic/loxodromic
- Choose disjoint Schottky disks
- Commutator of generators non-parabolic
- Limit set is a dust - disjoint, non-isolated points
- $0 < \dim_H(\Lambda) < 1$

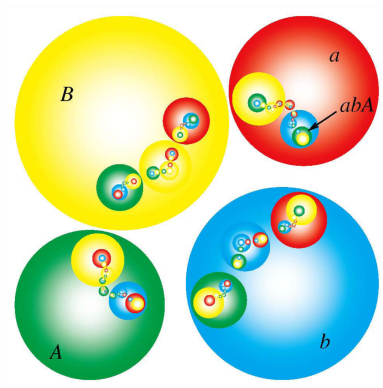


Image source: Indra's pearls - David Mumford et al. (p.106)

Fuchsian Schottky dust

- Choose all traces of maps to be real-valued
- Schottky group generated by hyperbolic maps
- Limit set is contained in circle orthogonal to all disks
- $M_1 = \csc \theta \begin{pmatrix} 1 & i \cos \theta \\ -i \cos \theta & 1 \end{pmatrix}$
- $M_2 = \csc \theta \begin{pmatrix} 1 & \cos \theta \\ \cos \theta & 1 \end{pmatrix}$
- $0 < \dim_H(\Lambda) < 1$

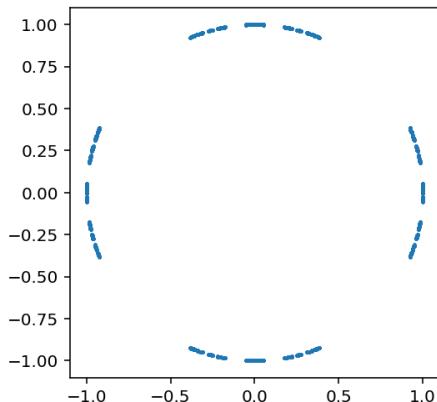


Figure: $\theta = 0.7$

Quasi-Fuchsian limit set

- Choose Schottky group with parabolic commutator
- Limit set is connected
- Constructed with real x, y and $k = (xy)^{-1} - \sqrt{(xy)^{-2} - 1}$
- $M_1 = \begin{pmatrix} \sqrt{1+x^2} & ikx \\ x/ik & \sqrt{1+x^2} \end{pmatrix}$
- $M_2 = \begin{pmatrix} \sqrt{1+y^2} & y \\ y & \sqrt{1+y^2} \end{pmatrix}$
- $1 \leq \dim_H(\Lambda) < 2$

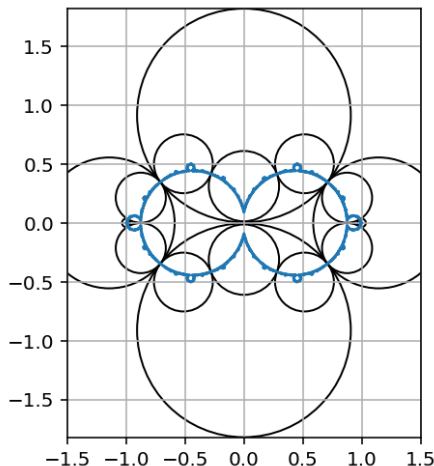


Figure: $x = 1.8, y = 0.1$

Fuchsian limit set

- Schottky group with parabolic commutator
- ensure traces of all maps are real
- Limit set is a circle
- Condition: $xy = 1, k = 1$
- $\dim_H(\Lambda) = 1$

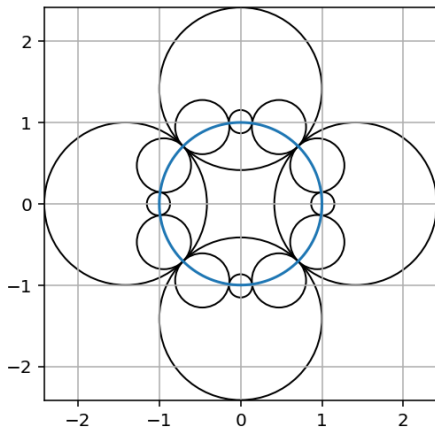


Figure: $x = y = 1$

Cusp groups

- Schottky group with parabolic commutator
- Choose either M_1 or M_2 to be parabolic
- Fixed points of M_i, M_i^{-1} are equal
- Resulting limit set features bulbs where fixed points lie
- formulae works with inputted traces of M_1, M_2 to construct generators
- $1 < \dim_H(\Lambda) < 2$

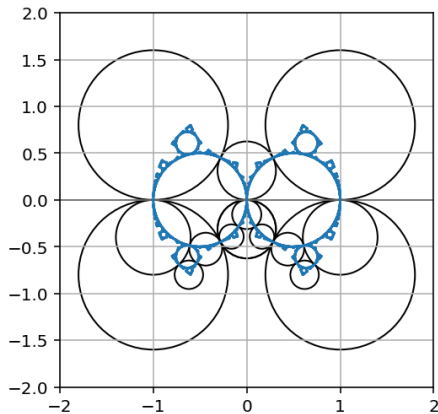


Figure: $\text{Tr}(M_1) = 2, \text{Tr}(M_2) = 2.5$

Double cusp groups

- Requires at least one generator to be parabolic
- Choose a second map to be parabolic
- Solve for the trace of second generator via polynomial
- Both generators parabolic creates Apollonian gasket with $\dim_H(\Lambda) \approx 1.3056$

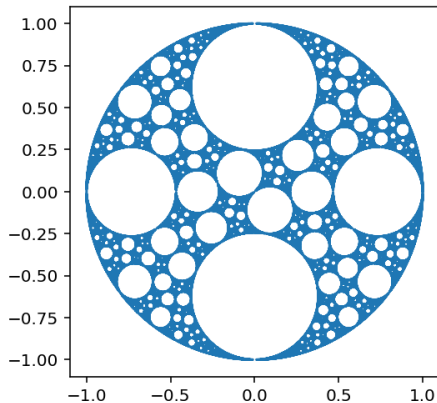


Figure: $\text{Tr}(M_1) \approx 1.64 - 0.76i$
 $\text{Tr}(M_2) = \text{Tr}(M_1^3 M_2^{-1} M_1^2 M_2^{-1}) = 2$

Conclusion

- Topics covered includes:
 - ▶ Real analysis
 - ▶ Complex analysis
 - ▶ Measure theory
 - ▶ Möbius map dynamics
 - ▶ Fractal analysis
- Utilised theory of Möbius maps to construct limit sets
- Provided a report for artistic and theoretical purposes